WaXAI: Explainable Anomaly Detection in Industrial Control Systems and Water Systems

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ABSTRACT

An Industrial Control System (ICS) plays a vital role in controlling and managing industrial processes. ICS predominantly operates without human supervision. This (mostly) autonomous role makes them an attractive target for adversaries. In recent years, machine learning (ML) algorithms have demonstrated their feasibility in detecting anomalies in sensor and actuator data, in an ICS. However, the ML algorithms demand extensive training time and lacks the ability to pinpoint the component(s) that are in an anomalous state. In this work, we employed two of the latest anomaly detection algorithms (ECOD and DeepSVDD) with a shorter training time, faster anomaly detection time, and a comparable efficiency rate in detecting anomalies. The algorithms were trained and tested using a dataset generated from the SWaT water treatment testbed. With the ubiquity of ML algorithms in decision making and forecasting, it is important for humans to perceive and understand its output decisions instead of viewing it as a black box oracle. In subsequent experiments, we employed eXplainable ML/AI (XML/XAI) models to explain the model's output decision, thus, increasing model transparency and trust. We also measure the effectiveness of the XAI models deployed thereby providing an indicator to which XAI models worked better in our experiments.

CCS CONCEPTS

• **Security and privacy → Intrusion/anomaly detection and malware mitigation**.

KEYWORDS

Industrial Control Systems Security, Critical Infrastructure Security, Anomaly detection, Artificial Intelligence, Explainable AI

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1 INTRODUCTION

An Industrial Control System (ICS) is predominantly an automated system that combines multiple types of control systems, such as supervisory control and data acquisition systems (SCADA), distributed control systems (DCS), and Programmable Logic Controllers (PLCs), which are typically encountered in industrial and critical infrastructures (CI) [28] such as water treatment plants, water distribution systems, and smart power grids. According to Industry 4.0 [30], cyber-physical systems, the Internet of Things (IoT), and cloud computing have all been incorporated into ICS to allow for increased autonomy and dependability.

Several anomaly detection algorithms have been developed to detect cyberattacks in ICS. Each of these approaches displays a successful prediction result, with the capacity to classify anomalous data. Many solutions using a Deep Neural Network (DNN)-based model demonstrate outstanding detection performance, but the principle disadvantage of this model type is its high computational cost. Researchers have emphasised that an anomaly detection model should have minimal hardware requirements and a short training period [11] to enable a prompt response to unexpected behaviour. There are recent unsupervised learning algorithms which have been developed to detect the outlier points showing excellent performance on several real-world datasets, such as Empirical-Cumulative-distribution-based Outlier Detection (ECOD) [19] and Deep Support Vector Data Description (DeepSVDD) [27]. Through experimentation, the authors noted that ECOD outperforms 11 popular baseline outline detector methods on 30 benchmarks. A few recent work such as COPOD [5] (based on ECOD) and an improved autoencoder that incorporates Deep SVDD in its training [31] has been applied to the ICS anomaly detection application. Our work adds to the recent investigations of ECOD and DeepSVDD but also includes model explainability in ICS. In Boateng and Bruce [5] and Wang et al. [31], XAI was not used to enhance model transparency, unlike in our work. In [5], a single model (autoencoder) was employed along with various XAI techniques for anomaly localisation, but the time taken for training XAI was not addressed. In contrast, our study employs two ML models and evaluates the training time of XAI, as well as identifies the optimal XAI for each model type.

While most of the previous research on ICS domains has focused on enhancing model performance, the anomaly detection models have remained hard to interpret. Since all decision support recommendations supplied by an AI based anomaly detector may not be accurate, it is important for the human operator to understand the AI's recommendation before arriving at a decision. This may be achieved by explaining the recommendation to the human operator, to assist with correct decision making. Further, ML algorithms used in AI models lacks the ability to localize anomalous occurrences in

the ICS. One way to address this issue is to employ eXplainable AI (XAI) models [2] in tandem with the anomaly detector.

Therefore, eXplainable Artificial Intelligence (XAI) plays a crucial role in model explanations and interpretations for understanding black-box models. Some existing anomaly detection algorithms has already integrated their model with XAI to increase transparency and reliability. However, past research has not measured the reliability of the XAI in ICS security, and as a result, there is a need to determine if one XAI approach is superior to another by measuring the quality of explanation — which refers to displaying the appropriate features for a certain anomalous event or anomaly localisation.

In order to provide high-quality interpretation for anomaly localisation, with low computational cost we apply recent algorithms to detect anomalous data in the ICS domain. Further, via measurements, we ascertain the best XAI method available for the water testbed use case. In this work, we utilised the Secure Water Treatment (SWaT) testbed dataset from Goh et al. [8] for training and testing two different anomaly detection models, ECOD and DeepSVDD, and compare their ability to localise anomalies using XAI models. We apply four different XAI techniques to explain the models' decisions — Shapley Additive Explanations (SHAP), Local Interpretable Model-agnostic Explanation (LIME), Accumulated Local Effects (ALE), and integrated gradients (IG). Further, we evaluate each XAI technique's accuracy score and computational time to determine the quality of XAI.

The remaining sections of this work are organised as follows: Section 2 provides relevant work on anomaly detection models in the ICS domain, followed by the description of the algorithms. The eXplainable AI approach utilised in this research is discuseed in section 3. A description of the SWaT dataset is provided in section 4. Section 5 demonstrates the methodology used in this research, beginning with the data pre-processing, how we set up and identify attack points from anomaly detection models, the XAI setup, and followed by a metric evaluation. Section 6 presents the results and discusses a case study, to explain how to interpret XAI results. We end with conclusions and details to further research and in section 7.

2 RELATED WORK

This section details related work on prior anomaly detectors in ICS, the models utilised in this paper — ECOD and DeepSVDD, and an overview of the XAI methodology used to explain the model's predictions.

Anomaly Detection. Several methods have been developed to prevent attacks on ICS systems, which are based on machine learning tasks that offer both supervised and unsupervised learning. Previous supervised learning algorithms from Kim et al. [13], used sequence-to-sequence neural networks to detect anomalous events. It uses time-series data from the SwaT dataset. The model uses a fixed length sliding window of length 100 seconds to capture the temporal context, and slides each window by 1 second). It took them a longer time to train (approximately 2 hours on the SWaT dataset). Other researchers have used an unsupervised learning method. For example, the work in Macas et al. [22] supplied an unsupervised framework for anomaly identification in water treatment systems. It applied a technique based on a Deep Neural Network (DNN) model, and with a high detection rate. The authors used the convolutional network (CNN) and Long-Short-Term Memory (LSTM) layers as attention-based layers to capture the dependencies of various variables (sensors and actuators). Similar to Macas et al. [22], the work in Li et al.[17] developed the Generative Adversarial Networks (GANs) model to capture the variable relationship based on the LSTM layer. They also compared their performance with other state-of-the-art unsupervised models such as PCA, One-Class SVM (OCSVM), K-Nearest Neighbor (KNN), Feature Bagging (FB), and Auto-Encoder (AE). We note that One-Class SVM (OCSVM), K-Nearest Neighbor (KNN), and Feature Bagging (FB) do not directly capture the temporal dependencies in multi-variate time series data, such as the one available in the SWaT dataset. Some of them, such as Inoue et al. [12] uses a fixed length sliding window to convert data into individual feature vectors. Consequentially, due to the fixed length of the window and/or ignoring other temporal dependencies, such methods are known to be susceptible to false positives (see Macas and Wu [22]).

Research from Inoue et al. [12] used unsupervised learning to compare the performance of the DNN model and OCSVM, demonstrating that the DNN model generated fewer false alarms while the OCSVM offers a slightly higher detection rate. The work from Kravchik et al. [15] compared the performance between CNN and LSTM models, demonstrating that the 1D CNN model is faster than the LSTM with a higher detection performance and lower false positive rate. The studies in Al-Abassi et al. [1] and Gomez et al. [9] uses semi-supervised learning techniques to detect attack scenarios of SWaT dataset, although due to the highly imbalanced dataset the work from Al-Abassi et al. [1] focuses on developing a new balance dataset using Stacked Autoencoder (SAE). The new balanced dataset was then passed to an ensemble model consisting of DNN and Decision Tree (DT) to detect cyber-attacks, resulting in high accuracy and a low false positive rate. In Gomez et al. [9], the authors focus primarily on developing feature engineering to increase detection performance. In Ïfahrmann et al. [7], the authors developed a lightweight LSTM-Variational Autoencoder (VAE) which resulted in an extremely lightweight model with a high detection rate. In Boateng and Bruce [5], the authors build a two-phase dual Copula-based Outlier Detection (COPOD) method. The first phase removes unwanted outliers using ECOD, and the second phase develops COPOD models based on the output data of the first phase. In Lachure and Doriya [16], they use a customised one-dimensional convolution neural network with k-fold cross-validation to detect anomalies in ICS.

ECOD and DeepSVDD. The work in Li et al. [19] proposed ECOD as an unsupervised learning approach. The ECOD has shown that it outperforms previous algorithms such as KNN, Isolation Forest (IForest), and PCA on 30 real-world benchmark datasets. In addition to efficiency and scalability, this approach has a time complexity of $O(n.d)$, where n is the total of data points and d is the number of dimensions. Deep Support Vector Data Description (DeepSVDD) is an unsupervised learning technique developed with the goal of anomaly detection by Ruff et al. [27]. DeepSVDD is based on kernel-based one-class classification and deep learning, employing a deep neural network to learn feature representation and map into the hypersphere.

XAI in ICS Security. Previous anomaly detection models in the ICS domain have demonstrated excellent accuracy for the detection rate, however, the prediction process is difficult to interpret due to the model's complexity. As a result, some recent works have integrated anomaly detection models with XAI models to increase the explainability and transparency of prediction. In Huong et al. [11], the authors developed VAE and SVDD models to detect attack scenarios and use XAI to explain how the models made their predictions, which is similar to the work from Ha et al. [10]. The work from Roshan and Zafar [26] demonstrated that XAI can be implemented not only for model explanation but also for feature selection, i.e., by selecting only the feature with a high importance value to train a new model and make predictions, leading to an improvement in the accuracy of predictions. In Ameli et al. [3], the author compares the effectiveness of multiple XAI models to localise anomalous occurrences in an attempt to demonstrate that XAI can also be used to identify anomalous events in ICS. Due to the nature of anomalous events being unknown and distinct from normal activity, we focused on an unsupervised learning method and employed the recent anomaly detection algorithms ECOD and DeepSVDD. Table 1 provides a brief description of recent research in the ICS security sector.

Work	Detection model	Data	Learning method	Model Runtime Assessment	XAI integration	Purpose of XAI	XAI Runtime Assessment	Year
[12]	LSTM, OCSVM	SWaT	Unsupervised	No	No	-	-	2017
[15]	LSTM, CNN	SWaT	Unsupervised	Yes	No	-	-	2018
[13]	LSTM	SWaT	Supervised	No	No	-	-	2019
[22]	Attention-based AE	SWaT	Unsupervised	No	No	-	-	2019
[17]	KNN, FB, PCA, EGAN, AE, MAD-GAN	SWaT, WADI	Unsupervised	No	No	-	-	2019
[1]	SAE, DNN, DT	SWaT	Semi-supervised	No	No	-	-	2020
[26]	AE	CICIDS2017 network dataset	Unsupervised	No	Yes	Feature selection	No	2021
[9]	LSTM	SWaT	Semi-supervised	No	No	-	-	2021
[7]	LSTM-VAE	SWaT, WADI	Unsupervised	Yes	No	-	-	2022
[10]	LSTM-AE, OCSVM	SCADA	Unsupervised	No	Yes	Model explanation	No	2022
[11]	VAE, SVDD	SCADA liquid storage infrastructure, SWaT	Unsupervised	Yes	Yes	Model explanation	No	2022
[3]	LSTM-AE	Server Machine Dataset	Unsupervised	No	Yes	Anomaly localisation	No	2022
[5]	COPOD	SWaT WADI TLIGHT	Unsupervised	Yes	No	-	-	2023
[16]	1D-CNN	WADI BATADAL	Unsupervised	Yes	No	-	-	2023
Ours	ECOD, DeepSVDD	SWaT	Unsupervised	Yes	Yes	Anomaly localisation	Yes	-

Table 1: Summary of recent works in ICS Security.

3 EXPLAINABLE AI

The purpose of XAI techniques is to develop machine-learning approaches that can provide understandable, reliable and explainable reasoning for black-box model predictions.

3.0.1 LIME. Local interpretable model-agnostic explanations (LIME) was developed by Ribeiro et al. [25]. To explain the individual predictions of black-box models, this technique utilizes the surrogate model, which is a simple model such as a linear, lasso, or decision tree. The surrogate models were trained to approximate specific predictions by producing a new dataset of perturbed samples. LIME then trains the surrogate model on the new dataset, which is weighted by the proximity of the sampled dataset to a certain prediction. The algorithm then weights additional samples based on their proximity to the explained instance. Then, a linear line is constructed to explain the data locally. The mathematical function of LIME can be expressed as follows:

$$explanation(x) = \arg\min_{g \in G} \quad L(f, g, \pi_x) + \Omega(g) \tag{1}$$

A model $g \in G$ where G is a class of possible interpretable models. The loss (e.g., mean squared error) is represented by $L(f, g, \pi_x)$), while the model complexity is represented by $\Omega(g)$ The proximity π_x specifies the size of the area surrounding sample x that we want to explain. To achieve both interpretability and local fidelity, one must minimize loss L while keeping $\Omega(g)$ as low as possible.

3.0.2 SHAP. SHapley Additive exPlanations (SHAP) by Lundberg and Lee [20] was developed to explain individual predictions based on the game theory by computing the contribution of each feature to the prediction. This approach calculates the Shapley values, which indicate how to distribute the prediction features, by iterating through all possible combination feature sets or coalitions. SHAP specifies the explanation as follows:

$$g(\acute{z}) = \Phi_0 + \sum_{j=1}^{M} \Phi_j \acute{z}_j \tag{2}$$

Where $g(\acute{z})$ is the explanation model, $\Phi_j \in R$ is the Shapley values for a feature j and each $\acute{z}_j$ is a simplified value of input features $\acute{z} \in \{0,1\}^M$, is the coalition vector of maximum coalition size M. Here, $\acute{z}_j = 1$ show the presence of the feature in the coalition, and its binary negation $\acute{z}_j = 0$ show the absence of the feature j. The Shapley value Φ_j can be calculated as follow:

$$\Phi j = \sum_{S \subseteq F} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_s(x_s)] \tag{3}$$

x_s denote the values of the input features in the feature subsets set S, where all $S \subseteq F$, where F is the set of all features. A model $f_{S \cup \{i\}}$ is trained with the feature present, whereas a second model f_s is trained without the feature. Then, the part $[f_{S \cup \{i\}}(x_{S \cup \{i\}}) - f_s(x_s)]$ is to compare the two models' prediction. Given that the effect of omitting a feature relies on other features, the preceding differences are calculated for all feasible subsets.

3.0.3 ALE. Previous most popular explanation techniques such as partial dependence plots (PDP) have weakness in producing error results when the data are correlated. This is due to it requiring extrapolation from the model results [23]). Hence, Accumulated Local Effects (ALE) was developed to overcome this shortcoming, and aimed at providing a lower computational cost, when compared with PDP (see Apley and Zhu [4]). The main objective of ALE is to visualize and understand the main effect on the set of features of each prediction result. ALE computes the difference in prediction conditional on the features and accumulates the divergence over the range of features, as shown in Equation 4:

$$\hat{f}_{S,ALE}(x_S) = \int_{z_{0,S}}^{x_s} \left(\int_{x_C} \hat{f}^S(z_s, X_c) d\mathbb{P}(X_C | X_S = z_S) d \right) dz_S - constant \tag{4}$$

where

$$\hat{f}^S(x_s, x_c) = \frac{\partial \hat{f}(x_S, x_C)}{\partial x_S} \tag{5}$$

Here, X_S is the set of features that we want to plot, X_C is other features in that we use to predict in model $\hat{f}$, S is the set of features where $s \in S$ and z_s is an interval where we calculate the changes. Regarding the marginal distribution of X_s, the constant attempts to either centre the plot vertically or decrease the mean of $\hat{f}^s$ to zero. Different intervals can be used to isolate the influence of the desired feature while limiting the impact of associated features. The features were divided into several intervals and the differences in the predictions were computed to evaluate the effects locally. This method is applicable to any model without derivatives using Equation 6:

$$\hat{f}_{j,ALE}(x) = \sum_{k=1}^{k_j(x)} \frac{1}{n_j(k)} \sum_{i:x_j^{(i)} \in N_j(k)} [\hat{f}(z_{k,j}, x_{-j}^{(i)}) - \hat{f}(z_{k-1,j}, x_{-j}^{(i)})] \tag{6}$$

$\hat{f}_{i,ALE}$ is an individual ALE explanation which is the sum of all individual instances. The right summation is the total effects of instances within an interval z_k where $k \in \{1, 2, ..., K\}$ while the left summation represents the aggregation of the average effects throughout all intervals. The neighborhood $N_i(k)$ is the total number of training instances within the kth interval. For defining the intervals, the quantiles of the feature's distribution are employed to provide equal data instances.

3.0.4 IG. Integrated Gradient was developed by Sundararajan et al. [29] to improve the prediction of DNN models based on the model output gradients relative to the input. This method is an implementation based on several existing XAI methods such as DeepLift and Layer-wise relevance propagation (LRP). The advantage over the prior approaches is that the basic deep neural network does not need to be modified, and can be applied to various data types not only the images but also text. This explanation was built on two fundamental axioms: sensitivity and implementation invariance. The satisfaction of sensitivity refers to the scenario where input and a baseline in one feature and the predictions are different, then a non-zero attribution is assigned to the different features. Similarly, implementation invariance means that for two similar functional networks, the attributions remain the same for the input and baseline instances. The IG can be calculated using Equation 7

$$IG_i(x) ::= (x_i - x_i') \times \int_{\alpha=0}^{1} \frac{\partial F(x' + \alpha \times (x - x'))}{\partial x_i} \tag{7}$$

F represents the DNN model where $F : R^n \rightarrow [0, 1]$, x is the input features and x' is the baseline input. The baseline refers to the zero vector feature for text and the black pixel for images. The gradients at all positions along the straight path from baseline to input are combined to compute IG value. The term $\frac{\partial F(x)}{\partial x_i}$ is the gradient along i^{th} dimension.

4 SWAT DATASET

The Secure Water Treatment (SWaT) testbed is a scaled-down version of a real-world industrial water treatment plant that provides a realistic sophisticated ICS environment. The complete explanation of the testbed and data set can be found in Goh et al. [8]. The SWaT dataset consists of normal plant operations and abnormal operations (when it is under various attacks). It is popularly by reseachers and practitioners for anomaly detection in a scaled down water treatment testbed for security testing and training — with over 4200 dataset requests from 89 regions since 2015.[1]

The following detail in this section is a short description of working procedures. Fig. 1 is the overview of SWaT testbed process consisting of 6 stages.

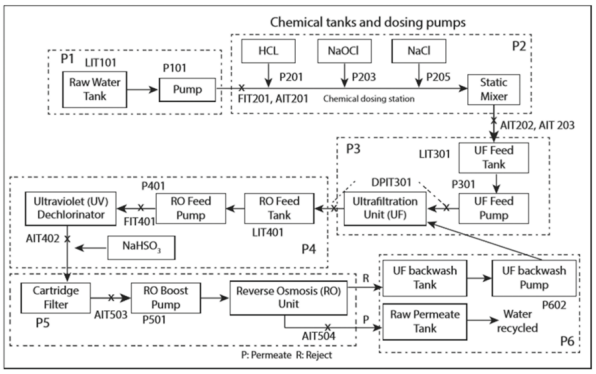


Figure 1: Overview of SWaT testbed [8].

In SWaT, the water flows through a six-step process from P1 to P6 and is disposed of in the final stage. Each stage has multiple sensors and actuators, including flow meters, level sensors, and pumps. The sensors and actuators of each process are shown in Table 2. The device names consist of two parts where the first 3 characters are shortened for devices type and the last 3 numbers refer to the process stage and number of devices. For example, the device FIT-101 refers to the flow meter in the first process where it is the first flow meter. This description of each device can be found as follows: FIT-XXX: Flow meter, LIT-XXX: Level Transmitter, AIT-XXX: Analyser, DPIT-XXX: Differential pressure indicating transmitter, PIT-XXX: Pressure meter, MV-XXX: Motorized valve, P-XXX: Dosing pump and UV-XXX: Dechlorinator.

Process	Sensor	Actuator
P1	LIT-101, FIT-101	MV-101, P101
P2	AIT-201, AIT-202, AIT-203, FIT-201	MV-201, P-201, P-202, P-203, P-204, P-205, P-206
P3	DPIT-301, FIT-301, LIT-301	MV-301, MV-302, MV-303, MV-304, P-301, P-302
P4	AIT-401, AIT-402, FIT-401, LIT-401	P-401, P-402, P-403, P-404, UV-401
P5	AIT-501, AIT-502, AIT-503, AIT-504, FIT-501, FIT-502, FIT-503, FIT-504, PIT-501, PIT-502, PIT-503	P-501, P-502
P6	FIT-601	P-601, P-602, P-603

Table 2: The sensors and actuators in each process.

The sensors and actuators at each stage are linked to a ring network (Level 0) that transmits data to the associated Programmable Logic Controllers (PLCs). The PLCs are connected to a star network (Level 1) to enable communication with Human Machine Interfaces (HMIs) and SCADA workstations.

[1] See https://itrust.sutd.edu.sg/itrust-labs_datasets/

The dataset was collected for 11 days and the first seven days contain normal working conditions. The last four days consist of anomaly data where 41 attacks were conducted. The plant was run from an empty state to a fully operational state. The attacks were conducted by altering the network traffic in the Level 1 network, spoofing the sensors' values, and issuing fake SCADA commands.

Attacker model considered in SWaT data collection. The adversary includes both internal and external attackers. Once inside the water plant's Operation Technology (OT), an attacker may carry out man in the middle (MITM) attacks to spoof sensor and actuator values viewed by the SCADA. Other attacks, such as on PLCs may be considered. However, the effect is to manipulate the view of the SCADA during data collection. The attack vector (the path the adversary chose to break into the plants OT network leads to four types of attacks). The four attack types are 1. Single Stage Single Point (SSSP) where exactly one sensor/actuator data is under attack, 2. Single Stage Multi-Point (SSMP) where multiple sensor/actuator data in exactly one stage is under attack, 3. Multistage Single Point (MSSP) where exactly one sensor/actuator data in multiple stages are under attack and 4. Multistage Multi-Point (MSMP) where multiple sensor/actuator data in multiple stages are under attack. The dataset consists of 946,722 records labeled either as an attack or normal, together with 51 attributes relating to sensor and actuator data. The attack data account for around 5.77% of the entire data.

5 METHODOLOGY

The working process in this study is shown in Figure 2. The SWaT dataset contains two sets of data. The first set of data contains only normal operation data, while the second set contains both normal and attack data. The required pre-processing using the historian dataset with the above two sets of data is carried out. We trained the models using the first dataset and made predictions on the second dataset. This study, utilised two different unsupervised models, ECOD and DeepSVDD for anomaly detection. Subsequently, we evaluated the models' predictions and applied XAI methods to explain the models' results on the detected anomaly scenarios as well as providing an evaluation of the XAI model (using Intersection Over Union, accuracy and computational time). These models were built using the PyOD library [34], a Python library intended for outlier identification and anomaly detection that is scalable and intuitive to use. The following sections describe the data preprocessing, anomaly detection model setup, XAI setup, and evaluation metric.

5.1 Data Pre-processing

This study primarily focused on attacks at the physical layer (sensor and actuator state) of the SWaT dataset. As a result, only physical layer data was considered. There are six processes with 51 features related to actuators and sensors. The training data obtained during regular behavior operations contains 496,800 records, while the test data contains both normal and attack events with a total of 449,919 records. No values are missing from the data. The 41 attacks that were initiated accounted for 5.77 % of total data, with attack duration ranging from 100 to 34,208 seconds. As indicated in Li et al. [18] and Xu et al. [32], the first 21,600 seconds (see Figure 3) were reduced due to the beginning of operation; hence, the adoption of this approach.

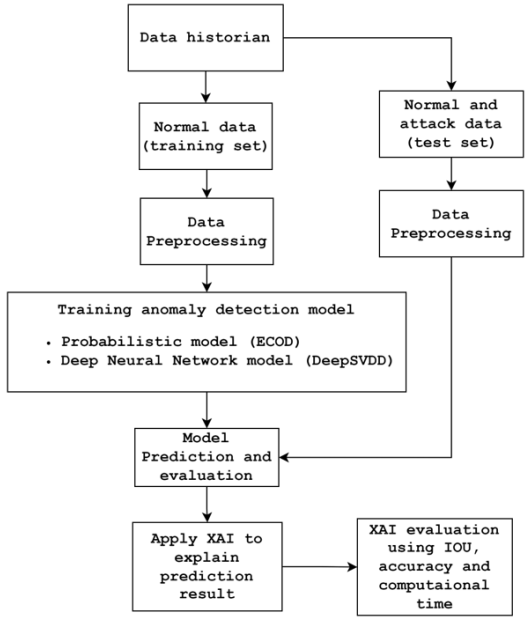


Figure 2: Overall methodology of the study.

Figure 3 depicts the value and distribution of the LIT-101 training set. Given that the distribution of characteristics does not correspond to the standard normal distribution, standardised data is not the best course of action. Instead, we normalise both continuous and discrete variables from the normal behaviour operation (training set) into a range of 0–1 and scaled the test data using the minimum and maximum values of the training data.

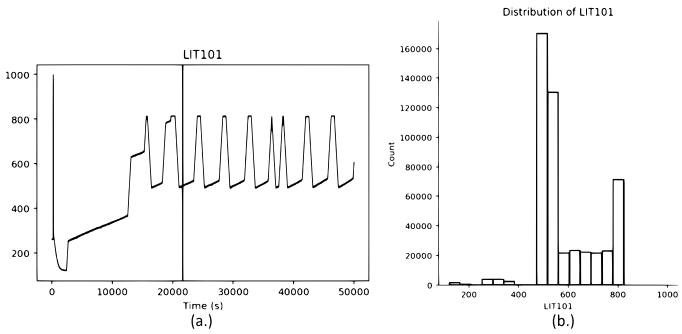


Figure 3: (a.) The value of LIT-101 during the starting process (b.) The x-axis is the level indicator measure and y-axis is the measure or time in seconds the overall value distribution of training set.

5.2 Model Set-Up, Training, and Prediction

5.2.1 Model Set-Up. This study employed two different anomaly detection model (ECOD and DeepSVDD), implemented[2] using the PyOD library. The ECOD has only 1 hyper-parameter, which is

[2]In expectation with reproducible experimentation [24], source code is available at https://github.com/Geebigib/Explainable-Anomaly-detection-In-ICS .

contamination (it refers to the amount of anomaly data in the training set). The contamination needs to be set in the range from 0 to 0.5 and due to our training set consisting of only normal data[3], we set at 1e-08 (tending to zero). The contamination is a mandatory value passed to the ECOD method in the PyOD library.

DeepSVDD is a neural network-based model with various hyper-parameters, including the size of the hidden neurons, the activation function, and the size of the training batch. Optionally, the model may be used as an AutoEncoder (AE), which will reverse the hidden neurons. To demonstrate, if we take two hidden neurons of sizes 64 and 32 and specify them using the AE type, the model architecture will consist of four neurons of sizes 64, 32, 32, and 64, respectively. Due to the presence of several parameters, we constructed five models with a different setup as shown in Table 3.

Model no.	AE type	Batch size	Hidden neuron size
1	Yes	64	(64, 32)
2	No	64	(64, 32)
3	No	64	(128, 64)
4	No	128	(64, 32)
5	No	128	(128, 64)

Table 3: The set-up of DeepSVDD model.

The process for parameter tuning is described as follows: We optimised model type by constructing two models, one with and one without AE type, and run them both on a training set. We optimised the batch size between 64 and 128. In accordance with the model size requirement that the size of the hidden neuron must be more than the total number of features, which is 51, hence sizes that exceed this range must be chosen. We adjusted two hidden neuron size pairs: (64,32) and (128,64).

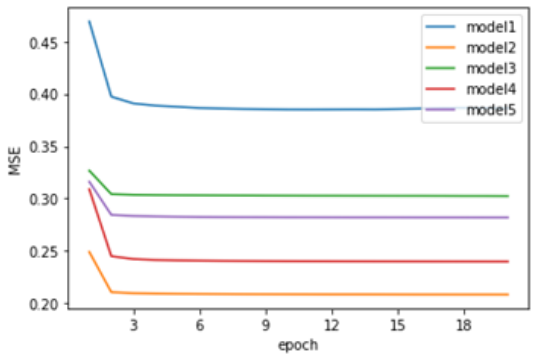


Figure 4: Validation loss of different setting of DeepSVDD.

As shown in Fig. 4. the validation loss of the model without AE type, is lower and performs faster than the AE type; consequently, we specified utilising the model without AE type while constructing the DeepSVDD model. We adjusted the DeepSVDD settings and selected those with the lowest validation loss which is model 2. The model was then constructed with the following parameters: 64 batch size, hidden neuron size (64,32), special model type without AE, and 100 epochs of training.

5.2.2 Model Training and Anomaly Decision. The single-stage technique was used to construct anomaly detection models, which can be accomplished by selecting only the sensors and actuators associated with a single process in order to train models and make predictions. We used model $n \in \{1, 2, \ldots, 6\}$ to learn the operating procedure of stage n and generate predictions on the stage's timestamp.

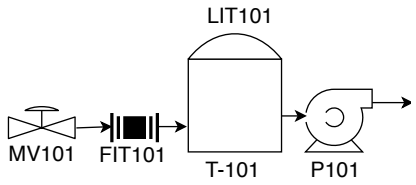


Figure 5: Detailed view of stage 1 of SWaT.

This process was chosen because it reduces the time and memory required to train independent process stages, as opposed to training all process stages into a single model. A more detailed view of stage 1 of SWaT is shown in Fig. 5. Fig. 6 illustrates the relationship between the flow meter (FIT-101) and level transmitter (LIT-101) in process P1, where it is obvious that the operating mechanisms of the sensor and actuator are connected. The FIT-101 monitors the raw water's flow rate, while the LIT-101 measures the tank's water level. When the water level in stage P1 reaches the maximum desired level, the valve (MV-101) will close, causing the FIT-101 to drop as depicted between 50,000 and 52,000 seconds. Therefore, it is reasonable to use a single-stage method for training and anomaly detection throughout the entire process.

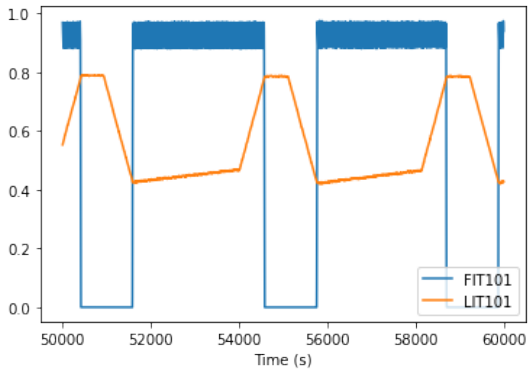


Figure 6: The relationship between flow meter (FIT-101) and level transmitter (LIT-101) process P1.

Both ECOD and DeepSVDD models give prediction results, as the error score of every single timestamp. According to *Li et al.* [19], the anomaly signifies the occurrence of unusual events. Therefore, we can infer that the greater the error score, the greater the possibility of an abnormality. Since the error scores of the training set and

[3]Training anomaly detection models solely on normal data allows them to learn the characteristic patterns of normal behavior effectively, facilitating improved anomaly detection during inference. Excluding anomaly data helps prevent bias towards anomalies and enables straightforward threshold setting based on the distribution of error scores. Additionally, models trained solely on normal data tend to generalise better to unseen data and are better interpretable, as abnormal instances are treated as deviations from the learned normal behavior.

test set of the two algorithms are in different ranges, the error score of the test set is normalised using the lowest and maximum values from the training set. Due to the training set comprising of normal operation records, scaling from the normal error score can also assist in reducing erroneous classification by diverting the test set's extreme error score. To decide the anomaly point, the threshold from the normalized error score must be set. If the threshold is inappropriately low, the model can detect more attack events, but this will result in a significant number of false alarms. False alarms can divert emergency responders from true crises and disrupt the devices' normal operation. Therefore, we must determine the appropriate thresholds that provide both a high possible attack detection rate and a low false alarm. Shown in Fig. 7 is the distribution of error score on the test set of process P1.

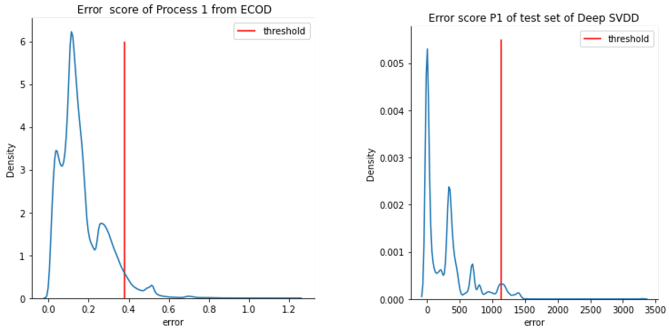


Figure 7: Distribution of error score on the test set of ECOD (a.) and DeepSVDD (b.) of process P1.

The normalised error score of the test set of process P1 was used for the threshold setting with the GHOST algorithm [6]. The GHOST algorithm is aimed to modify the threshold that yields the optimal number for the classification of imbalanced data. We selected only stage P1 because we wanted the model to be capable of reacting to unforeseen attacks and provide excellent detection performance. Using a threshold at 95 percentiles of error score for both models will result in the lowest number of false alarms and the highest detection rate. After the thresholds were selected, a decision on whether the records are anomalies was made by comparing the error score with the threshold at each stage. If the score of one or more stages within one timestamp exceeds the threshold, we labeled the entire timestamp as attack records. If the timestamps between i and $i + 100$ seconds are labeled as attacks, we defined the attacks as having been detected. We considered 100 more timestamps in an attempt to minimise the number of false alarms[4]. We only show one process to drive the intuition, but it may be used as a metric to decide how and when an output is determined to be a detected attack.

5.3 XAI Set-Up

This study employed four XAI techniques, including kernel SHAP, LIME, ALE, and IG. To evaluate the effectiveness of XAI in terms of explanation accuracy, we employed the IOU (Intersection Over Union) similarity and XAI accuracy metrics discussed in section 5.4.1. Before applying the anomaly score with XAI, we sample 100 records from all detected attacks.

Due to the attack duration ranging from 100 to 34,208 seconds, implementing XAI for prolonged attacks will be time and resource-expensive. After sampling the attack records, the XAI is employed to explain the feature contribution score derived from the model's decision for every sensor and actuator because we would like to assess the quality of explanation throughout the entire process which can indicate the ability to localise attacked devices. Traditionally, LIME is applied with a local explanation or a single instance; hence, we employed the Submodular pick (SP-LIME) algorithm, which provides the explanation result as a representative set of feature scores across instances. For IG explanation, we implemented this approach only with the DeepSVDD model, as it requires the construction of DNN models using the PyTorch or Keras frameworks while others are applied with both models. For SHAP explanation, we applied the algorithm from Lundberg et al. [20] and the visualization from Lundberg et al. [21]. The algorithm from Ribeiro et al. [25] is used for LIME explanation and Alibi from Klaise et al. [14] for ALE and IG explanations.

5.4 Evaluation Metric

5.4.1 Anomaly Detection Evaluation Metric. To evaluate the performance of two different anomaly detection models, at each stage, the metrics appropriate for imbalance classification problems were calculated: precision (Equation 8), recall (Equation 9), and F1-score (Equation 10). In addition, we measured performance by comparing the total number of attacks detected and the time consumption for each process in TABLE 4. The True Positive (TP) is the record that is accurately classified as a normal class. True Negative (TN) is the attack record that is labeled as abnormal, whereas False Positive (FP) is an attack record that has not been detected. False Negative (FN) is the false alarm that occurs when normal data is misclassified as an attack. For performance evaluation, we also plotted the Receiver Operating Characteristics (ROC) curve, which is a conventional metric indicating the model's efficiency over all feasible thresholds in the range of True Positive Rate (TPR) or recall and False Positive Rate (FPR) (Equation 11). We then calculated the area under the curve (AUC) from the ROC curve to determine the ability to differentiate between normal and attack classes. The greater the AUC, the more accurately the model can recognise anomalous records.

$$Precision\ (P) = \frac{TP}{TP + FP} \tag{8}$$

$$Recall\ (R)\ or\ True\ Positive\ Rate = \frac{TP}{TP + FN} \tag{9}$$

$$F1 - score = \frac{2 \times R \times P}{R + P} \tag{10}$$

$$False\ Positive\ Rate = \frac{FP}{FP + TN} \tag{11}$$

5.4.2 XAI Evaluation Metric. Since we intended to evaluate the quality of explanations for four unique XAI techniques over the attack events, we adopt the metric proposed by Ameli et al. [3] for comparing XAI quality. The positive score indicates that the

[4]Note, we employ a 95th percentile threshold coupled with a criterion where the error score must exceed this threshold for at least 100 seconds to detect anomalies. This duration aligns with the minimum length of attacks in our dataset, reducing false alarms by focusing on sustained deviations from normal behavior.

feature is a significant factor in the prediction, while the negative value indicates that the feature is unimportant. We selected only the positive score because it indicates the feature that is crucial to the model's choice and can refer to compromised attacked devices. Intersection over Union (IOU) similarity (Equation 12) and simple Accuracy coefficients (Equation 13) are used to measure the performance of XAI, where y' is the set of positive contribution score of XAI at the given anomaly points, y is the ground truth feature, which in this case is the attack sensors or actuators, and K is the total number of attacks detected. We also analyse the computational time over 100 sample explanations to identify the technique that offers the highest explanation quality in terms of the ability to identify an attacked device with the fastest explanation result.

$$Intersection\ over\ Union\ (IOU) = \frac{1}{K}\sum\left(\frac{|\{y' \wedge y\}|}{|\{y' \vee y\}|}\right) \tag{12}$$

$$Accuracy = \frac{1}{K}\sum\left(\frac{|\{y' \wedge y\}|}{|y'|}\right) \tag{13}$$

In several attack occurrences, only one device is attacked; hence, the number of positive feature contribution scores may exceed the number of true attack features. Therefore, we applied the top k features with positive contributions scores from Yang et al. [33] who use the top-k hit ratio to improve the recommendation system's accuracy. This method was also adopted in Ameli et al. [3]. In this work, the top 5 and 10 features are used to calculate IOU similarity, whereas all positive features are used to calculate accuracy. This can help generate a more reliable performance evaluation. We selected the top 5 and top 10 features because, when an attack occurs, we will investigate a subset of the features that contribute to the abnormality first. This is because human observers can only comprehend small chunks of data. Therefore, by utilising the top features, the domain expert would be able to simply track the location of the attack. Therefore, the XAI model should display the appropriate top features to accurately identify an abnormality. To illustrate the calculation, assume that the attack device is LIT-101, which refers to the ground truth feature y. The top five features with a positive score, according to the explanation result, are P-102, MV-101, LIT-101, AIT-202, and FIT-401. The top 5 features IOU similarity is 1/5 and the accuracy is 1.

6 RESULT AND DISCUSSION

6.1 Model Performance Comparison

We compared the performance of ECOD and DeepSVDD which are probabilistic and DNN models respectively. The models predict the anomaly score for each SWaT process based on a single timestamp. The greater the anomaly score, the greater the risk of attack events. As illustrated in Figure 8, the ROC curves for each process in SWaT are computed from the raw anomaly score for every feasible threshold. After plotting the ROC curve, the AUC was computed for each process to determine its ability to distinguish between normal and abnormal records. The AUC can be found on Table 4 as well as the precision, recall, F1-score, computation time and the total attack detected from the models. In several processes, both ECOD and DeepSVDD achieved a high AUC score, but for process P6 of both models, the AUC is poor. This is because there is only one attack in TABLE 5 related to the actuator P-602 (see TABLE 2) in this stage.

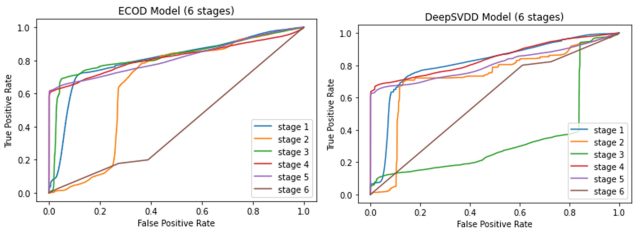


Figure 8: The ROC curve of process P1 to P6 from the (a.) ECOD and (b.) DeepSVDD.

The entire performance is calculated from the final prediction for which we set a threshold, and if one or more of the six stages have an anomaly score greater than the threshold, we declare that the system is under attack. Of the 41 attacks in dataset, only 36 of them have a physical impact. Hence, only those 36 attacks (shown in TABLE 5) are considered. Both models have comparable performance in terms of detection rate, while ECOD achieves slightly higher recall at 0.75, F1 score at 0.74, and total attacks detected at 30 out of 36, or an approximate general detection rate of 83.33 percent. For DeepSVDD, the model can detect 29 attacks with a detection rate of 80.56 percent and a precision rate of 0.76. In contrast to other processes, P2 also achieved poor performance with low precision, recall, and F1-score. This is because there are only four attack occurrences (attacks 6, 24, 29, and 30) while ECOD failed to detect *attack* 6 from this stage, with DeepSVDD failing to detect *attack* 29.

Evidently, the ECOD uses little training and prediction time at every stage, with the fastest training and prediction time occurring in P6. This is because P6 consists of only four sensors and actuators, whereas P2's model required the most time due to its 11 functional devices. We trained the DeepSVDD model for 100 epochs, so the approximate training time for each stage is 24 minutes or 14 seconds per epoch. Because the ECOD model uses a probabilistic distribution function to find anomalies and is parameter-free, it can be considered to use simpler algorithms than the DNN model. Therefore, it is reasonable for ECOD to require less computational time than the complex DeepSVDD model (see TABLE 4)[5]. This implies that the sensors and actuators of each process are highly correlated to one another and that the attacks diverge from the normal operating condition; therefore, by training on each process, the models have a high detection rate.

We identified false alarms based on the final anomaly detection decision that predicts whether the system is in a normal state or under attack. By utilising all data from the first process to set the threshold and then selecting the threshold with the highest F1-score using the GHOST algorithm, we attempted to minimise false alarms and true negatives, which indicate that attacks were forecast as normal operation stages. The confusion matrix from the models is presented in Figure 9. The ECOD achieves a greater true negative (attack records were accurately predicted) at 6.96% of

[5]ECOD training time in Table 4 sums to a little over 9 seconds and not 14 seconds as expected. The difference arises from the inclusion of latency during stage transitions, which contributes to the overall training time from stage 1 to 6 totaling 14 seconds. Conversely, measuring each stage individually resulted in shorter duration. This also happened with DeepSVDD, albeit with a longer duration, making it less noticeable.

Model	Stage	Metric						
		Precision	Recall	F1 score	AUC	Training time (sec)	Prediction time (sec)	Total attack detected
ECOD	P1	0.59	0.58	0.59	0.805	1.57	3.11	
	P2	0.46	0.48	0.46	0.657	2.23	4.62	
	P3	0.78	0.69	0.73	0.824	1.84	3.75	
	P4	0.97	0.75	0.81	0.818	1.40	2.10	**30**
	P5	0.96	0.72	0.79	0.816	1.59	3.19	
	P6	0.49	0.50	0.48	0.413	0.41	0.79	
	Overall	0.73	**0.75**	**0.74**	-	-	-	
DeepSVDD	P1	0.65	0.53	0.52	0.813	1,433.47	15.88	
	P2	0.48	0.50	0.47	0.728	1,366.34	16.12	
	P3	0.62	0.52	0.52	0.346	1,388.32	16.01	
	P4	0.94	0.58	0.61	0.846	1,410.62	16.45	29
	P5	0.94	0.58	0.61	0.807	1,474.61	16.31	
	P6	0.49	0.50	0.48	0.587	1,523.24	16.19	
	Overall	**0.76**	0.69	0.72	-	-	-	

Table 4: Model performance in each stage and the total attack detected.

the total prediction, but DeepSVDD is able to detect fewer attack records at 5.1%. The DeepSVDD offered slightly better performance in avoiding false alarms, but a larger rate of false negatives at 7.02%, whereas the ECOD false negatives are only at 5.18%. This could be due to the fact that the DeepSVDD anomaly score is more distinguishable for anomaly records. Consequently, when the threshold is set to the 95th percentile of the distribution of anomaly score, the normal operation records will be more localised in the low score zone, resulting in fewer false alarms. Although some records are incorrectly classified, both models demonstrated the capacity to classify normal operation and attack events.

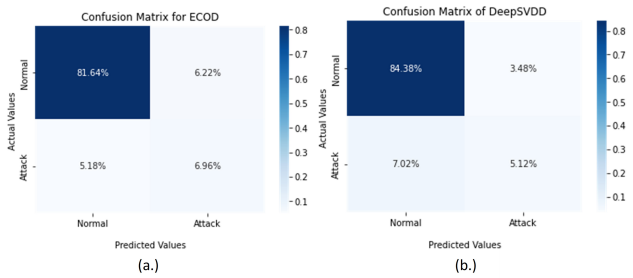


Figure 9: Confusion matrix of (a.) ECOD and (b.) DeepSVDD.

6.2 Analysis

According to the results, ECOD and DeepSVDD cannot detect *attack* 6 and *attack* 7, respectively. We investigate the reason why the attacks were not detected, given that the anomaly scores while the system is under attack are easily distinguishable from those of a functional system. As illustrated in Figure 10, the ECOD is unable to detect *attack* number 6, which happens on AIT-202. The normal operating condition of the device is greater than 7.05, but when an attack occurs, it decreases to 6, showing sudden drops. Although the predicted anomaly scores were unusually high when the attack occurred, it was undetected because the anomaly scores are lower than the threshold. This is the consequence of minimising false alarms. Unlike *attack* 6, the other attacks have been identified because their anomaly scores are more distinct from normal conditions, such as *attack* 1. The instability of this attack's anomaly scores can be related to the discrete operating mechanism (on/off) of MV-101, which in the attack is triggered by a state change of the valve from CLOSE to OPEN.

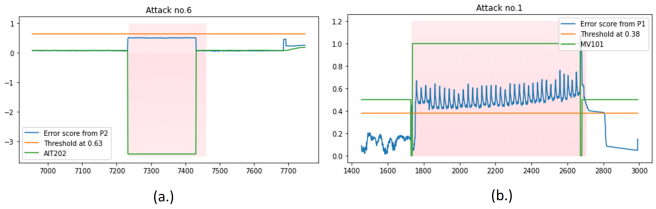


Figure 10: The anomaly scores from the process under attacks, (a.) set threshold and attacked device corresponding to *attack* number 6, (b.) *attack* number 1.

As displayed in Figure. 11 (a.), *attack* 14, which had not been detected by the ECOD model, was further examined. This attack occurred on MV-303, which was triggered to remain closed while it ought to be open. We can see that after a prolonged period of attacks, the anomaly score suddenly rises. This can occur when the operation of another device is disrupted, causing the model to indicate slightly higher scores for anomalies at subsequent timestamps but are lower than the threshold. The failure to detect this attack was a result of the working mechanism of MV303 which appeared to be functioning properly during the early phase of the attack, although there is also a possibility that the attack failed because the tank in process P3 was already full. As shown in Fig. 11 (b.), attack no.16 occurred on LIT-301, which was manipulated to drop by 1 millimeter per second without any actual system change. This threat is undetectable by the ECOD model because the anomaly scores increase gradually as the sensor's value is changed. Therefore, the detection duration is less than 100 seconds, as our method classifies attacks as detected if the anomaly scores are more significant than the threshold for 100 seconds to reduce the false alarm rate. Another case study demonstrates that we were unable to detect an attack using the DeepSVDD model due to the wide distribution of anomaly scores and the extreme value of the threshold at 95 percentiles. As shown in Fig. 11 (c.), the attack period of attack no.22 against UV-401, AIT-502, and P-501 is shown. We plotted the process P5 anomaly score to determine the undetected cause. The threshold of P5 is enormously high; accordingly, we are unable to detect it due to the wide distribution of anomaly scores from the model, even though the anomaly scores rose when the system was under attack. There are additional case studies in which the models predict a low anomaly score, but if an attack occurs for which we cannot determine the cause, we describe these attacks as undetectable.

In conclusion, we attempted to identify the root cause of undetected attacks and discovered four major reasons. The explanation behind this reasoning is as follows:

- The anomaly score is high for the attack period but lower than the threshold because we want to minimise false alarms by setting the threshold high.
- The working mechanism of the attacked device appears normal, and there is a probability that the attack may fail. I.e., despite the attack, the outcome are a set of normal operational states.
- The attack period anomaly score is higher than the threshold but less than the 100 seconds set to reduce false alarm rate.

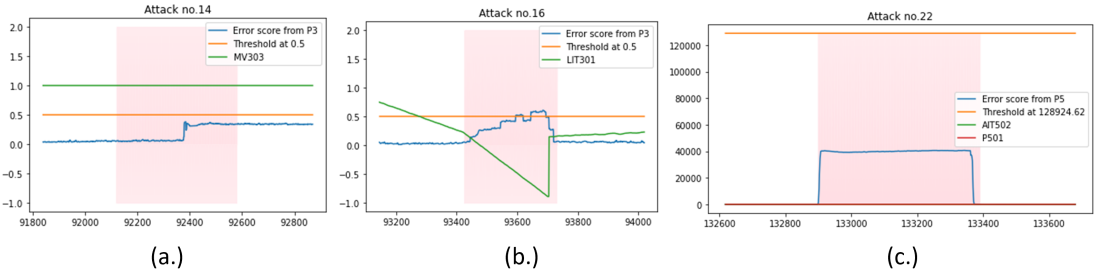


Figure 11: The anomaly scores from the process under attacks, threshold, and attacked device of (a.) *attack* 14 , (b.) *attack* 16 and (c.) *attack* 22.

- The models are unable to detect the attack resulting from a low anomaly score (undetectable).

As illustrated in Table 5, there were 41 attacks of which 36 that had a physical impact on SWaT. Only those attacks that had a physical impact are displayed. Table 5 for SWaT describes the attacked devices, the attack scenario, whether the attacks are detected or not by the models, and the reason for the undetected attack.

Attack no.	Attack point	Attack event	ECOD	DeepSVDD	Reason for undetected attacks
1	MV-101	Open MV-101	Yes	Yes	-
2	P-102	Turn on P-102	Yes	Yes	-
3	LIT-101	Increase by 1 mm every second	Yes	Yes	-
4	MV-504	Open MV-504	Yes	-	Threshold is extremely high
6	AIT-202	Set value of AIT-202 as 6	-	Yes	Anomaly score below threshold due to minimizing false alarm
7	LIT-301	Water level increased above High-High set point	-	Yes	Unable to detect
8	DPIT-301	Set value of DPIT as >40kpa	Yes	Yes	-
10	FIT-40	Set value of FIT-401 as <0.7	-	-	Unable to detect
11	FIT-401	Set value of FIT-401 as 0	Yes	Yes	-
13	MV-304	Close MV-304	Yes	-	Threshold is extremely high
14	MV-303	Do not let MV-303 open	-	Yes	MV-303 seems to be in normal condition and the attack is likely failed
16	LIT-301	Decrease water level by 1mm each second	-	Yes	Detection length less than 100 sec
17	MV-303	Do not let MV-303 open	Yes	Yes	-
19	AIT-504	Set value of AIT-504 to 16 μS/cm	Yes	Yes	-
20	AIT-504	Set value of AIT-504 to 255 μS/cm	-	-	Unable to detect
21	MV-101, LIT-101	Keep MV-101 on continuously; Value of LIT-101 set as 700 mm	Yes	Yes	-
22	UV-401, AIT-502, P-501	Stop UV-401, Value of AIT502 set as 150, Force P-50 to remain on	Yes	-	Threshold is extremely high
23	P-602, DIT-301, MV-302	Value of DPIT-301 set to >0.4 bar, Keep MV-302 open, Keep P-602 closed	Yes	Yes	-
24	P-203, P-205	Turn of P-203 and P-205	Yes	Yes	-
25	LIT-401, P-401	Set value of LIT-401 as 1000, P402 is kept on	Yes	Yes	-
26	P-101, LIT-301	P-101 is turned on continuously, Set value of LIT-301 as 801 mm	Yes	Yes	-
27	P-302, LIT-401	Keep P-302 on continuously Value of LIT401 set as 600 mm till 1:26:01	Yes	Yes	-
28	P-302	Close P-302	Yes	Yes	-
29	P-201, P-203, P-205	Turn on P-201, Turn on P-203, Turn on P-205	Yes	-	Threshold is extremely high
30	LIT-101, P-101, MV-201	Turn P-101 on continuously; Turn MV-101 on continuously; Set value of LIT-101 as 700 mm; P-102 started itself because LIT301level became low	Yes	Yes	-
31	LIT-401	Set LIT-401 to less than L	Yes	Yes	-
32	LIT-301	Set LIT-301 to above HH	Yes	Yes	-
33	LIT-101	Set LIT-101 to above H	Yes	Yes	-
34	P-101	Turn P-101off	Yes	Yes	-
35	P-101, P-102	Turn P-101 off, Keep P-102 off	Yes	Yes	-
36	LIT-101	Set LIT-101 to less than LL	Yes	Yes	-
37	P-501, FIT-502	Close P-501, Set value of FIT-502 to 1.29 at 11:18:36	Yes	Yes	-
38	AIT-402, AIT-502	Set value of AIT402 as 260, Set value of AIT502 to 260	Yes	Yes	-
39	FIT-401, AIT-502	Set value of FIT-401as 0.5, Set value of AIT-502 as 140 mV	Yes	-	Threshold is extremely high
40	FIT-401	Set value of FIT-401 as 0	Yes	Yes	-
41	LIT-301	decrease value by 0.5 mm per second	Yes	Yes	-

Table 5: The detected attacks from each model and the reasons for undetected attacks.

6.3 XAI Performance Comparison

We evaluated the quality of XAI models based on their ability to locate an attacked device using top5 and top10 IOU similarity, accuracy coefficient, and computational time. We applied the anomaly score of the 30 and 29 attacks detected by ECOD and DeepSVDD, respectively. Due to the varying attack lengths, we sample 100 records from the detected attack[6] to perform with XAI and then ranked the positive contribution score to compute the IOU's similarity and accuracy; the findings are presented in Table 6.

XAI Approach	ECOD				Deep SVDD			
	IOU top 5	IOU top 10	Acc.	Time (sec)	IOU top 5	IOU top 10	Acc.	Time (sec)
SHAP	6.67%	6.57%	87.77%	1,369.21	6.89%	5.86%	82.76%	53.51
LIME	5.33%	5.00%	53.33%	3,527.59	4.14%	3.08%	37.93%	2,266.60
ALE	3.46%	5.26%	30.27%	854.85	4.83%	2.41%	24.13%	7.31
IG	-	-	-	-	3.45%	3.79%	58.62%	5.33

Table 6: The ability of XAI to localise attacked devices and the computation time over 100 samples. See section 6.3 for the explanation of poor IOU.

The experiment demonstrates that the SHAP method achieved the highest IOU similarity and accuracy score for both ECOD and DeepSVDD results. This could be because SHAP uses game theory which is simpler and directly connects with anomaly score results to generate the features contribution score. The method iterates over multiple potential features to determine their value resulting in higher time consumption for ECOD. The computing time of SHAP for ECOD is greater than 22 minutes, but DeepSVDD used less than 1 minute. The difference in computational time is due to the ECOD using probability distributions to produce the anomaly score that was required to re-calculate for every explanation, whereas the weights of the DeepSVDD model were assigned from the training model and fixed in the prediction process. When the XAI iterated over the probabilistic model, it needed to calculate the probability distribution of each feature on a single data, which took longer to compute when compared to the DNN model. LIME required the longest computational time of the two models. This may be because we adopted SP-LIME, which represents a single collection of feature scores derived from all data and is a method that requires the construction of a surrogate model for each record, resulting in a significant computational time. The IG method requires the least computing time for the DeepSVDD model since it is

[6]We used the criteria that prediction errors must consistently exceed the designated threshold for a minimum duration of 100 seconds to be classified as an attack An attack is classified as detected only when it passes this criteria. The selection of the 100-second threshold aligns with the observed minimum duration of attacks within the SWAT dataset. Due to the attack duration ranging from 100 to 34,208 seconds, implementing XAI for prolonged attacks will be time and resource-expensive.

optimised for usage with the DNN model, where gradients are used to generate the feature contribution score. Although IG had a strong performance for short computing time, this method is restricted to gradient-based models only. Consequently, we cannot apply the ECOD model. The ALE method has the second-fastest processing time since it calculates the difference in prediction conditional on the features and is therefore less complex than SHAP and LIME.

Explanation for poor IOU. The top 5 and top 10 IOU similarities of each model are poor since there are only one or two attacked devices relevant to the attack scenario as an overall process explanation consisting of 51 features, thus, resulting in poor localisation performance. This would indicate that if XAI were applied to identify the location of anomalous devices, it may need to be applied within the process, which is contrary to our current approach. By applying XAI to a single process, the model can capture more information about the interaction between sensors and actuators in the process, resulting in more precise localisation of anomalies. The majority of the top 5 IOU similarity scores are more significant than the top 10 indicating that the XAI approach can precisely identify the attacked devices due to the high contribution score of the attacked features in the top 5 positive scores which is related to the high accuracy score.

6.4 XAI Result Visualisation and Interpretation

The XAI algorithm explains the prediction outcome by displaying the contribution of features to the model's decision. In this part, we explain how to interpret visualised XAI results. Attack number 24 occurred in process P2, where the P-203 and P-205 are attacked by a physical change from open to closed and that ECOD and DeepSVDD can detect the attack. We used the result of DeepSVDD anomaly scores to explain four XAI and sampled 100 data from the attack period to identify the root cause. Figure. 12 displays the feature score of (a.) SHAP, (b.) the average feature score from IG, (c.) the feature score from LIME, and (d.) the ALE plots of attacked features.

The SHAP summary plots illustrate the overall feature importance and attacking feature effects. The X-axis represents the Shapley value for each record of the feature, while the Y-axis indicates the importance of the feature. Recall that attack number 24 occurred in process P2, where the P-203 and P-205 are attacked by a physical change from open to closed. With a Shapley value range of -0.2 to 0.1, the SHAP identifies MV-201 as the most critical feature of the attack, indicating that the model considers MV-201 to be the most significant element of attacked devices. This contradicts the actual attack point, which could be the result of employing the entire process explanation and bias from sample attacked data. The second most significant attribute is P-203, which relates to the attack devices while P-205 is among the top ten anomaly scores that influenced the model's result. If another attack occurs we can utilise this information to identify the location of the attack by examining the relevance of the top features.

The explanations in Fig. 12 localises the anomalies by informing the plant operator where to look based on the measure of the features scored by the XAI. The XAI methods demonstrate that P-203 is the most crucial characteristic for model decisions. The IG feature contribution score is derived from the sample data's average feature score. In this attack, the P-203 is given the greatest importance score compared to devices. This provides the same result as LIME, which provides the greatest positive score for feature P-203. The ALE plots of P-203 further indicate that when this feature's value changes, it influences the model's decision, as represented by the slope line over the possible score range of -1 to 1. IG, LIME, and ALE plots have not demonstrated the significance of P-205. As a result of the attempt to explain the entire process, there may be unrelated features in the prediction score, resulting in a lack of accuracy in displaying actual feature importance.

7 CONCLUSIONS

Anomaly detection is one of the primary security mechanisms used in ICS to prevent devastating losses and impacts on human life. In this paper, two outlier detection models, ECOD and DeepSVDD, with the SWaT dataset were employed to compare the detection performance and time consumed in training and testing. We trained the model on multiple SWaT processes to allow for understanding of the relationship between sensors and actuators that are controlled by PLCs during the process. Both models can detect more than 80% of attack scenarios. While ECOD outperforms DeepSVDD in terms of the F1-score and total number of attacks detected, DeepSVDD has a higher precision score, indicating a lower false alarm rate. Because ECOD is based on probabilities, training and prediction times are approximately 5 times faster than DeepSVDD. Although the models demonstrated the ability to detect the attack events, their false alarm rates are significant.

The models were trained, with predicted anomaly scores for each SWaT timestamp; however, it is difficult to explain why the algorithms predicted a given score. Consequently, we implemented the result with XAI to enhance model transparency and acquire confidence in the model's decisions. Since multiple XAI had been developed, we compared their performance in terms of their ability to locate attacked devices and their computational time. By examining the feature score given by XAI models, the domain expert can discover anomalous devices to help combat a cyberattack. To compare the performance of XAI, we calculate the feature score for SHAP, LIME, ALE, and IG, using the anomaly score derived from the anomaly detection models. It was discovered that SHAP performs the best at locating the anomalous while IG demonstrates the fastest computational time for the DNN model and LIME expends the most computing time when compared to the other approaches. In addition, we found that the computational time of XAI models differs from that of anomaly detection models.

Future work: We look to provide a human interpretable reason why the AI believes the reported anomaly to be an anomaly. This is to assist the operator further understand the decision recommendation made by the AI. Further, we will look to provide a safety score by measuring the severity (w.r.t. plant safety) of the anomaly/anomalies detected, given the required information. The safety score may be used by the operator to assist with the decision whether a plant emergency shutdown initiation is required, or if the plant is safe to continue to run in the presence of detected anomalies, while further diagnostics are carried out. It is also of interest to investigate the relationship between model complexity

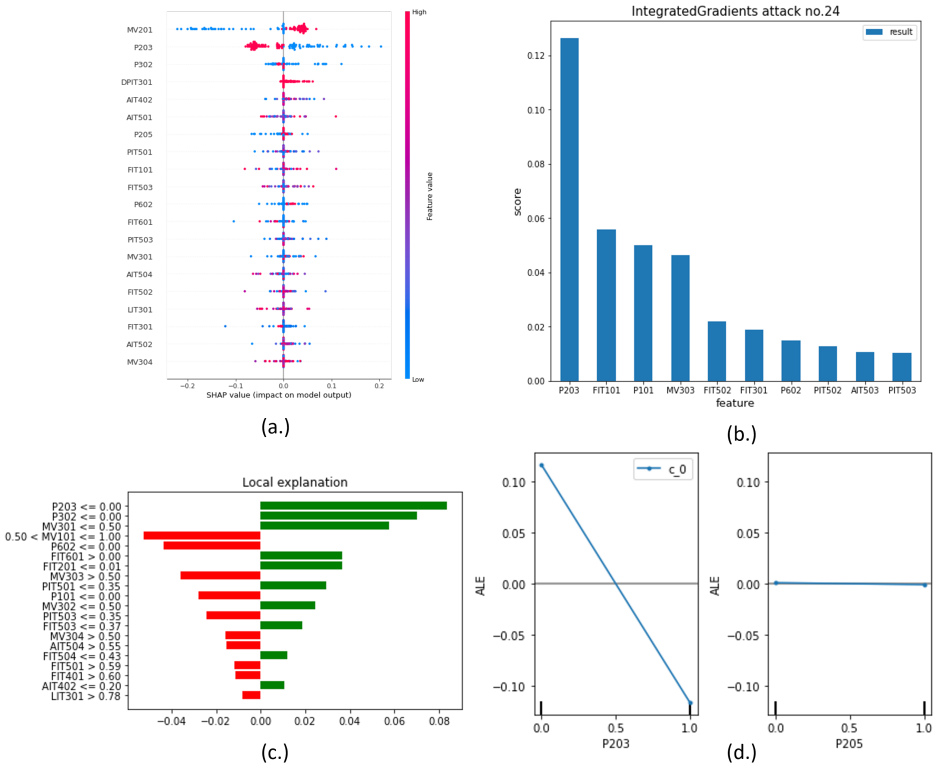


Figure 12: The explanation from (a.) SHAP, (b.) IG, (c.) LIME, and (d.) ALE plots.

and XAI to determine the optimal XAI approach for each model type.

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